IfThenEval: Measuring the Inconsistency of Conditional Instruction Following in Large Language Models

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Abstract

Large language models are increasingly deployed with system prompts containing conditional (if-then) rules, yet no dedicated benchmark evaluates how reliably models follow such instructions. We introduce IFTHENEVAL, a benchmark of 120 test cases spanning 8 conditional instruction types—keyword, format, content, length, negation, nested, conjunction, and disjunction—each with balanced trigger and non-trigger inputs and deterministic verification. We evaluate GPT-4.1 and GPT-4O-MINI and find that conditional instruction-following capacity varies dramatically by instruction type: GPT-4.1 ranges from 100% accuracy on keyword conditions to 75% on nested conditionals, while GPT-4O-MINI ranges from 100% to 50%. Critically, the difficulty ranking across instruction types is highly consistent between models (Spearman $\rho = 0.83$, $p = 0.011$), suggesting that certain conditional instruction types are inherently harder rather than reflecting model-specific weaknesses. Error analysis reveals an asymmetry in failure modes: the stronger model predominantly over-triggers (false positives), while the weaker model predominantly misses triggers (false negatives). We establish a difficulty hierarchy—keyword $<$ negation $<$ disjunction $<$ format $<$ conjunction $<$ content $<$ length $<$ nested—and show that the capacity bottleneck lies in the complexity of individual conditions, not the number of simultaneous conditions. Our findings provide actionable guidance for practitioners designing conditional system prompts.

1 Introduction

When deploying large language models (LLMs) as assistants and agents, practitioners routinely embed conditional rules in system prompts: "if the user asks about pricing, respond with the latest rate card"; "if the input contains profanity, decline to answer." These *if-then instructions* govern when and how the model should act, and their reliability directly determines the trustworthiness of the deployed system. Yet when such rules misfire—triggering when they should not or failing to trigger when they should—the consequences range from poor user experience to safety violations.

A growing body of work evaluates LLM instruction following, including IFEval [Zhou et al., 2023], FollowBench [Jiang et al., 2023], and COLLIE [Yao et al., 2023]. These benchmarks focus on *imperative* instructions ("write in JSON", "keep it under 100 words") but do not isolate the *conditional* aspect: whether the model correctly determines when a rule applies. The SIFo benchmark [Chen et al., 2024b] includes a security-rules task that is implicitly conditional, and Heo et al. [2024] show that internal instruction-following signals do not generalize across instruction types—hinting that conditional capacity may vary systematically. However, no benchmark directly and comprehensively evaluates if-then instruction following.

We fill this gap with IFTHENEVAL, the first systematic benchmark for conditional instruction following. We design 120 test cases spanning 8 instruction types that cover a range of cognitive demands: simple keyword matching, format switching, content detection, length control, negation,

nested branching, conjunction, and disjunction. Each test case pairs a conditional instruction with an input that either triggers or does not trigger the condition, and compliance is verified by deterministic checker functions inspired by IFEval's verifiable-instruction paradigm [Zhou et al., 2023]. We evaluate two models from the GPT family—GPT-4.1 and GPT-4O-MINI—and analyze accuracy, false positive rates, and false negative rates per instruction type.

Our main findings are:

- We introduce IFTHENEVAL, a benchmark of 8 conditional instruction types with balanced trigger/non-trigger inputs and deterministic verification, filling a gap in instruction-following evaluation.
- We show that if-then capacity varies dramatically by instruction type (25–50 percentage point ranges) and that the difficulty ranking is consistent across models (Spearman $\rho = 0.83$, $p = 0.011$), establishing an inherent hierarchy of conditional instruction difficulty.
- We identify an asymmetry in failure modes: stronger models predominantly over-trigger (false positives) while weaker models predominantly miss triggers (false negatives), providing distinct design implications for practitioners.
- We demonstrate that the capacity bottleneck lies in the complexity of individual conditions rather than the number of simultaneous conditions, with both models maintaining near-perfect accuracy on up to 4 concurrent simple conditions.

The rest of this paper is organized as follows. Section 2 reviews related work on instruction-following evaluation. Section 3 describes the IFTHENEVAL benchmark and evaluation protocol. Section 4 presents our main results and analysis. Section 5 discusses implications and limitations. Section 6 concludes.

2 Related Work

Instruction-following evaluation. The evaluation of LLM instruction following has received considerable attention. IFEval [Zhou et al., 2023] introduced the concept of *verifiable instructions*—constraints that can be checked automatically—and defined 25 instruction types covering keywords, length, format, and content constraints. FollowBench [Jiang et al., 2023] extended this with multi-level constraint difficulty, showing that models degrade as constraints become more complex. COLLIE [Yao et al., 2023] proposed a grammar-based framework for compositional constraint specification at word, sentence, paragraph, and passage levels. Qin et al. [2024] studied how performance degrades with multiple constraints composed together. These benchmarks focus on *imperative* instructions ("always do X") and do not isolate the conditional reasoning required by if-then instructions. Our work specifically targets the conditional dimension.

Sequential and compositional instruction following. The SIFo benchmark [Chen et al., 2024b] evaluates sequential instruction following, where each step depends on the previous one. Its security-rules task—which requires following conditional access rules—is the closest existing proxy for conditional instruction evaluation. However, it covers only one narrow task type and does not systematically vary the type of conditional logic. CoDI-Eval [Chen et al., 2024a] evaluates controllability under diversified constraint expressions, showing significant gaps between open-source and closed-source models. We build on these ideas by creating a taxonomy of 8 conditional instruction types and measuring performance across all of them.

Internal mechanisms of instruction following. Heo et al. [2024] discovered an "instruction-following dimension" in LLM embedding space that predicts compliance but, critically, does not generalize across instruction types. This finding directly motivates our hypothesis: if internal signals for instruction following are type-specific, then conditional instruction-following capacity should vary systematically across types. Our empirical results confirm this prediction.

Improving instruction following. Several recent works propose methods to improve instruction adherence. Kumar et al. [2025] showed that expressing instructions as pseudo-code yields 8–21% gains on instruction-following benchmarks, suggesting that structured representations help models parse conditional logic. Wu et al. [2025] found that chain-of-thought reasoning can sometimes *hurt* instruction following, particularly for constraint adherence. Zhang et al. [2024] demonstrated that generalization to unseen instructions requires diverse training data across semantic domains.

Table 1: The 8 conditional instruction types in IFTHENEVAL. Each type has 2–3 instruction templates, and each template has 3 trigger and 3 non-trigger inputs (12–18 cases per type, 120 total).

Type	Condition	Example instruction	Cases
KEYWORD	Input contains keyword	"If 'urgent' appears, start with PRIORITY:"	18
FORMAT	Input is about topic	"If about a person, respond in JSON"	18
CONTENT	Input mentions concept	"If a color is mentioned, include hex code"	18
LENGTH	Input has meta-property	"If single word, answer in one sentence"	12
NEGATION	Input does NOT have X	"If no question mark, respond in uppercase"	18
NESTED	If A, then: if B do X; else Y	"If beverage: if hot, 'Enjoy warm!'; if cold, 'Best chilled!' "	12
MULTI-AND	Both A and B present	"If both '?' and 'why', start with 'Great question!' "	12
MULTI-OR	Either A or B present	"If 'help' or 'assist', start with 'I'm here to help!' "	12

These findings inform how practitioners might improve conditional instruction reliability beyond benchmark measurement.

3 Methodology

3.1 Benchmark Design

We construct IFTHENEVAL, a benchmark of 120 test cases for evaluating conditional instruction following. Each test case consists of three components: (1) a *conditional instruction* placed in the system prompt, (2) a *user input* that either satisfies or does not satisfy the condition, and (3) a *deterministic verifier* that checks whether the model's response correctly follows (or correctly does not follow) the conditional action.

Instruction types. We define 8 types of conditional instructions, summarized in table 1. The types span a range of cognitive demands: simple string matching (KEYWORD), output format switching (FORMAT), semantic content detection (CONTENT), meta-property reasoning (LENGTH), logical negation (NEGATION), multi-level branching (NESTED), conjunction (MULTI-AND), and disjunction (MULTI-OR).

Balanced trigger design. For each instruction template, we create an equal number of *trigger* inputs (condition should activate) and *non-trigger* inputs (condition should not activate). This 50/50 balance allows us to separately measure false positives (over-triggering on non-trigger inputs) and false negatives (missing triggers on trigger inputs), providing a more fine-grained view than overall accuracy alone.

Deterministic verification. Each instruction type has a corresponding verifier function that checks the model's response against the expected behavior. For example, the KEYWORD verifier checks whether the response starts with the required prefix when the keyword is present and does not start with the prefix when the keyword is absent. Verifiers include tolerance for minor formatting variations (e.g., trailing whitespace, case variations where appropriate).

3.2 Scaling Benchmark

To test how performance changes with the number of simultaneous conditions, we create a separate scaling benchmark of 23 test cases. Each case contains 1, 2, or 4 keyword-to-marker conditional instructions in a single system prompt (e.g., "if input contains 'alpha', include [ALPHA] in response; if input contains 'beta', include [BETA]"). We evaluate both overall accuracy (all conditions correct) and per-marker accuracy (fraction of individual conditions correctly handled).

3.3 Models and Configuration

We evaluate two models from OpenAI's GPT family:

- GPT-4.1: a state-of-the-art instruction-following model.
- GPT-4O-MINI: a smaller, faster model for capability comparison.

Table 2: Overall accuracy on IFTHENEVAL (120 test cases). Both models substantially exceed the 50% random baseline, but a significant gap remains between the two models.

Model	Accuracy (%)	95% CI
GPT-4.1	**94.2**	[88.4, 97.1]
GPT-4O-MINI	82.5	[74.7, 88.3]
Random baseline	50.0	—

Both models are queried via the OpenAI API with temperature $= 0.0$ and seed $= 42$ for reproducibility. Maximum generation length is 500 tokens, sufficient for all response types.

3.4 Evaluation Metrics

We report the following metrics for each instruction type and model:

- **Accuracy**: fraction of test cases where the model's response is correct (action taken when triggered, no action when not triggered).
- **True positive rate** (TPR): fraction of trigger inputs where the conditional action is correctly performed.
- **True negative rate** (TNR): fraction of non-trigger inputs where the conditional action is correctly withheld.
- **False positive rate** (FPR $= 1 -$ TNR): fraction of non-trigger inputs where the model incorrectly performs the action.
- **False negative rate** (FNR $= 1 -$ TPR): fraction of trigger inputs where the model fails to perform the action.

3.5 Statistical Analysis

We use Pearson's chi-squared test for independence to assess whether accuracy varies significantly across instruction types within each model. We compute Cramér's V as an effect size measure. To test whether difficulty rankings are consistent across models, we compute the Spearman rank correlation ρ between per-type accuracy vectors. We report 95% confidence intervals computed via the Wilson score method. All tests use a significance level of $\alpha = 0.05$.

4 Results

4.1 Overall Performance

Table 2 shows overall accuracy on the main benchmark. GPT-4.1 achieves 94.2% accuracy (95% CI: [88.4%, 97.1%]), substantially outperforming GPT-4O-MINI at 82.5% (95% CI: [74.7%, 88.3%]). Both models are well above the 50% random baseline, confirming that LLMs do possess meaningful conditional instruction-following capacity—but the gap between models and the wide confidence intervals suggest considerable room for improvement.

4.2 Accuracy by Instruction Type

Table 3 and figure 1 present accuracy broken down by instruction type. The variation is striking. GPT-4.1 achieves perfect scores on 4 of 8 types (KEYWORD, FORMAT, NEGATION, MULTI-OR) but drops to 75.0% on NESTED conditionals—a 25 percentage point range. GPT-4O-MINI shows an even wider 50 percentage point range, from 100% (KEYWORD) to 50.0% (LENGTH).

For GPT-4O-MINI, the variation across types is statistically significant ($\chi^2 = 20.72$, $df = 7$, $p = 0.004$, Cramér's $V = 0.42$). For GPT-4.1, the variation approaches significance ($\chi^2 = 13.30$, $df = 7$, $p = 0.065$, Cramér's $V = 0.33$), likely limited by the smaller number of errors.

Table 3: Accuracy (%) by instruction type. Types are sorted by average accuracy across both models. Best result per type in **bold**. All types show GPT-4.1 $\geq$ GPT-4O-MINI, but the gap varies from 0 (KEYWORD) to 41.7 percentage points (LENGTH).

Type	GPT-4.1	GPT-4O-MINI	Avg.
KEYWORD	**100.0**	**100.0**	100.0
NEGATION	**100.0**	94.4	97.2
MULTI-OR	**100.0**	91.7	95.8
FORMAT	**100.0**	88.9	94.4
MULTI-AND	**91.7**	83.3	87.5
CONTENT	**88.9**	77.8	83.3
LENGTH	**91.7**	50.0	70.8
NESTED	**75.0**	58.3	66.7

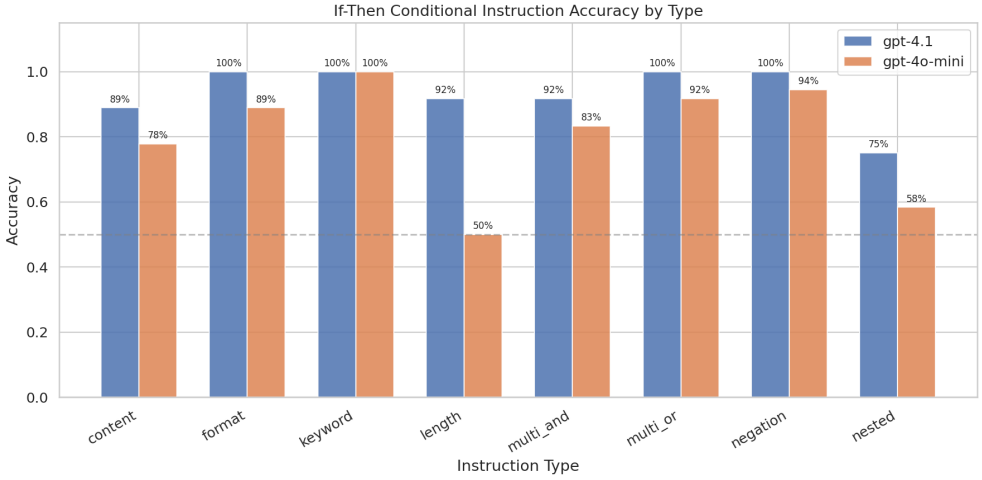


Figure 1: Accuracy by conditional instruction type for both models. GPT-4.1 (dark) achieves perfect accuracy on 4 types but drops notably on NESTED and CONTENT. GPT-4O-MINI (light) shows a wider range, with LENGTH and NESTED being particularly challenging. The difficulty ordering is largely consistent across models.

4.3 Cross-Model Difficulty Consistency

Figure 2 shows the per-type accuracy for GPT-4.1 versus GPT-4O-MINI. The Spearman rank correlation is $\rho = 0.83$ ($p = 0.011$), indicating a strong and statistically significant agreement in difficulty ranking. Both models find KEYWORD easiest and NESTED hardest. This consistency suggests that the difficulty of conditional instruction types is an inherent property of the task rather than a model-specific weakness.

4.4 Error Analysis: False Positives vs. False Negatives

Table 4 breaks down errors into false positive rates (FPR, over-triggering) and false negative rates (FNR, under-triggering) for each model and instruction type. The two models exhibit strikingly different failure modes.

GPT-4.1 predominantly over-triggers. GPT-4.1's errors are concentrated in false positives: it performs the conditional action even when the condition is not met. The worst case is NESTED (FPR $= 33.3\%$), followed by CONTENT (FPR $= 22.2\%$) and MULTI-AND (FPR $= 16.7\%$). The CONTENT false positives reveal semantic over-generalization—for example, when asked to include a hex color code only if a color is mentioned, the model included `#228B22` (forest green) for an input about plants, associating the topic with the color green.

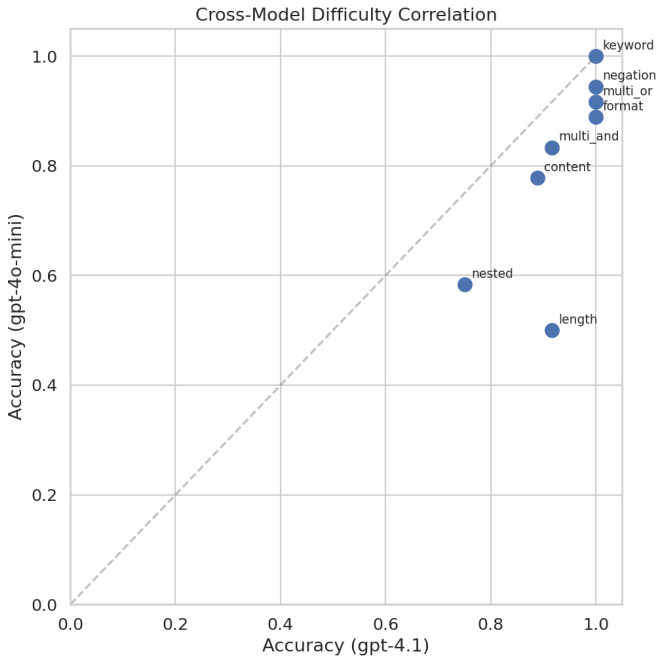
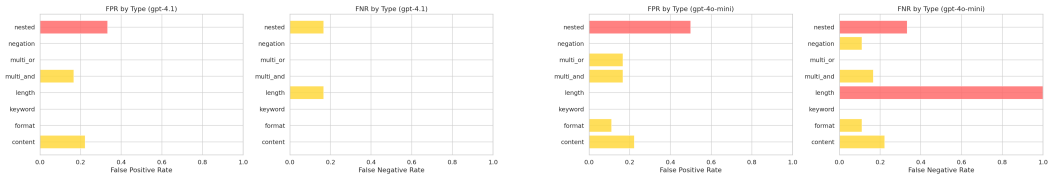


Figure 2: Cross-model accuracy scatter plot. Each point is one instruction type. The strong positive correlation (Spearman $\rho = 0.83$, $p = 0.011$) indicates that the difficulty ranking of conditional instruction types is consistent across models.



(a) GPT-4.1

(b) GPT-4O-MINI

Figure 3: Error type breakdown by instruction type. *(a)* GPT-4.1 errors are dominated by false positives (over-triggering), especially for NESTED and CONTENT. *(b)* GPT-4O-MINI errors are dominated by false negatives (under-triggering), with LENGTH showing complete failure (FNR $= 100\%$).

GPT-4O-MINI predominantly under-triggers. GPT-4O-MINI's errors are concentrated in false negatives: it misses conditions that should fire. Most dramatically, LENGTH conditions have FNR $= 100\%$—the model never correctly adjusts response length based on input properties. This is a consistent capability gap, not a stochastic failure. NESTED conditions also show high FNR (33.3%) alongside high FPR (50.0%), indicating general confusion rather than a directional bias.

4.5 Scaling with Multiple Conditions

Table 5 shows results from the scaling benchmark. Both models maintain near-perfect accuracy even with 4 simultaneous keyword-to-marker conditions. GPT-4.1 achieves 91.7% overall accuracy with 2 conditions and recovers to 100% with 4 conditions; GPT-4O-MINI achieves 100% throughout. Per-marker accuracy (individual condition compliance within multi-condition prompts) remains $\geq$95.8% for both models.

This result contrasts with the dramatic variation in the main benchmark. The key difference is that the scaling benchmark uses simple keyword-to-marker conditions—the easiest type. The capacity

Table 4: False positive rate (FPR) and false negative rate (FNR) by instruction type. GPT-4.1 errors skew toward false positives (over-triggering), while GPT-4O-MINI errors skew toward false negatives (under-triggering). Highest error rates per model in **bold**.

	GPT-4.1		**GPT-4O-MINI**	
Type	FPR(%)	FNR(%)	FPR(%)	FNR(%)
KEYWORD	0.0	0.0	0.0	0.0
FORMAT	0.0	0.0	11.1	11.1
NEGATION	0.0	0.0	0.0	11.1
MULTI-OR	0.0	0.0	16.7	0.0
MULTI-AND	16.7	0.0	16.7	16.7
LENGTH	0.0	16.7	0.0	**100.0**
CONTENT	22.2	0.0	22.2	22.2
NESTED	**33.3**	16.7	**50.0**	33.3

Table 5: Scaling benchmark results. Overall accuracy requires all conditions in a prompt to be correct; per-marker accuracy measures individual condition compliance. Both models handle up to 4 simple concurrent conditions with minimal degradation.

	1 condition	**2 conditions**	**4 conditions**
Overall accuracy (%)			
GPT-4.1	**100.0**	91.7	**100.0**
GPT-4O-MINI	**100.0**	**100.0**	**100.0**
Per-marker accuracy (%)			
GPT-4.1	—	95.8	**100.0**
GPT-4O-MINI	—	**100.0**	**100.0**

bottleneck is in the *complexity* of individual conditions, not the *number* of concurrent conditions (at least up to 4 simple conditions).

4.6 Difficulty Hierarchy

Combining all results, we establish the following hierarchy of conditional instruction difficulty (averaged across both models, from easiest to hardest):

1. KEYWORD (100.0%): Simple string-matching conditions are trivially reliable.
2. NEGATION (97.2%): "If NOT X" conditions are handled surprisingly well.
3. MULTI-OR (95.8%): Disjunctive conditions are near-perfect.
4. FORMAT (94.4%): Format-switching on topic detection works well.
5. MULTI-AND (87.5%): Conjunctive conditions show some degradation.
6. CONTENT (83.3%): Semantic content detection is harder than keyword matching.
7. LENGTH (70.8%): Length control based on input properties is unreliable.
8. NESTED (66.7%): Multi-level conditional branching is the hardest.

5 Discussion

Why do some types fail? The difficulty hierarchy reveals a pattern: conditions that require *semantic reasoning* about the input (content detection, length assessment, nested branching) are harder than conditions that require *surface-level pattern matching* (keyword presence, negation of keyword presence). This aligns with the finding of Heo et al. [2024] that instruction-following signals in model embeddings do not generalize across instruction types—different types likely engage different internal mechanisms.

The complete failure of GPT-4O-MINI on LENGTH conditions (FNR $= 100\%$) is particularly striking. The model never correctly shortens or lengthens its response based on input properties such as word count. This suggests that smaller models do not attend to *meta-properties* of the input (e.g., "is this a single word?") when those properties serve as conditions. Larger models handle this better but still imperfectly (91.7%).

False positive vs. false negative asymmetry. The finding that stronger models tend to over-trigger while weaker models tend to under-trigger has a natural interpretation. GPT-4.1, with greater capacity, is more sensitive to potential triggers and more eager to apply conditional rules—even when the condition is not strictly met. This manifests as semantic over-generalization (e.g., associating "plants" with the color green). GPT-4O-MINI, with less capacity, is more conservative and misses genuine triggers. This asymmetry echoes the precision–recall tradeoff in classification: stronger models sacrifice precision for recall, while weaker models do the opposite.

Implications for practitioners. Our results yield actionable guidance for designing conditional system prompts:

- **Prefer keyword-based conditions** over semantic or length-based conditions whenever possible. Keyword conditions are perfectly reliable across both models.
- **Avoid nested if-then-else logic.** Flatten nested conditions into separate, independent rules.
- **Test with both trigger and non-trigger inputs.** Testing only positive cases will miss the over-triggering failure mode that predominates in stronger models.
- **Expect different failure modes by model size.** When using smaller models, watch for missed triggers; when using larger models, watch for over-triggering.

Implications for research. Our findings suggest that conditional instruction following is a distinct capability from imperative instruction following and warrants dedicated evaluation. The consistent difficulty hierarchy across models ($\rho = 0.83$) implies that improving conditional instruction following may require targeted interventions—possibly structured instruction representations such as pseudo-code [Kumar et al., 2025] or specialized training on diverse conditional types [Zhang et al., 2024].

Limitations. We acknowledge several limitations of this work. *Sample size*: With 120 test cases (12–18 per type), our per-type estimates have moderate statistical power. Larger samples would yield tighter confidence intervals. The marginal significance of the chi-squared test for GPT-4.1 ($p = 0.065$) may reflect this limitation. *Model coverage*: We test only two models from the same family (OpenAI GPT). Evaluating models from other families (Claude, Gemini, Llama, Qwen) is needed to validate whether the difficulty hierarchy generalizes beyond the GPT family. *Condition complexity*: Our conditions are relatively clean and unambiguous by design. Real-world conditional instructions are often vaguely specified, and performance on natural conditions may be substantially worse. *Scaling benchmark scope*: The scaling experiment uses only simple keyword-to-marker conditions. Scaling with complex condition types composed together may show different patterns. *Single run*: While temperature 0.0 provides deterministic outputs, we do not capture variance under stochastic sampling conditions.

6 Conclusion

We introduced IFTHENEVAL, the first benchmark specifically designed to evaluate conditional (if-then) instruction following in large language models. Across 120 test cases spanning 8 conditional instruction types, we found that if-then capacity is significantly inconsistent: accuracy ranges from 100% on keyword conditions to as low as 50% on length conditions depending on the model. The difficulty ranking is highly consistent across models (Spearman $\rho = 0.83$, $p = 0.011$), estab-

lishing a hierarchy of conditional instruction difficulty: keyword < negation < disjunction < format < conjunction < content < length < nested. We further showed that the capacity bottleneck lies in the complexity of individual conditions rather than the number of concurrent conditions, and that stronger and weaker models exhibit opposite failure modes—over-triggering versus under-triggering, respectively.

These findings have immediate practical implications: practitioners should prefer keyword-based conditions, avoid nested logic, and test prompts with both trigger and non-trigger inputs. For the research community, our results motivate dedicated benchmarks and training methods for conditional instruction following as a distinct capability.

Future work should expand IFTHENEVAL to more models and model families, increase sample sizes, test complex multi-type condition compositions, and explore whether structured instruction formats or targeted fine-tuning can close the gap on hard condition types like nested branching and length control.

References

Xintong Chen et al. CoDI-Eval: Benchmarking LLMs on controllable generation under diversified instructions. *arXiv preprint*, 2024a.

Xinyi Chen, Baohao Liao, Jirui Qi, Panagiotis Eustratiadis, Christof Monz, Arianna Bisazza, and Maarten de Rijke. The SIFo benchmark: Investigating the sequential instruction following ability of large language models. *arXiv preprint arXiv:2406.19999*, 2024b.

Juyeon Heo, Christina Noh, Jiwon An, Seonghyeon Lee, and Sohee Park. Do LLMs "know" internally when they follow instructions? *arXiv preprint arXiv:2410.14516*, 2024.

Yuxin Jiang, Yufei Wang, Xingshan Zeng, Wanjun Zhong, Liangyou Li, Fei Mi, Lifeng Shang, Xin Jiang, Qun Liu, and Wei Wang. FollowBench: A multi-level fine-grained constraints following benchmark for large language models. *arXiv preprint arXiv:2310.07641*, 2023.

Aditya Kumar, Srinivas Murthy, Sumit Bhat, and Danish Contractor. Training with pseudo-code for instruction following. *arXiv preprint*, 2025.

Yiwei Qin et al. Benchmarking complex instruction-following with multiple constraints composition. *arXiv preprint*, 2024.

Yufei Wu et al. When thinking fails: Pitfalls of reasoning for instruction-following. *arXiv preprint*, 2025.

Shunyu Yao, Howard Chen, Austin W. Hanjie, Runzhe Yang, and Karthik Narasimhan. COLLIE: Systematic construction of constrained text generation tasks. *arXiv preprint arXiv:2307.06439*, 2023.

Karel Zhang, Yixin Wang, and François Charton. Only-IF: Revealing the decisive effect of instruction diversity on generalization. *arXiv preprint*, 2024.

Jeffrey Zhou, Tianjian Lu, Swaroop Mishra, Siddhartha Brahma, Sujoy Basu, Yi Luan, Denny Zhou, and Le Hou. Instruction-following evaluation for large language models. *arXiv preprint arXiv:2311.07911*, 2023.